Sandilya Kambhampati

Fishers, Indiana
463-206-9053 | kprsandilya@gmail.com | linkedin.com/in/kprsandilya/ | github.com/kprsandilya

Education

Purdue University **August 2026 – May 2028**
M.S. in Computer Science *West Lafayette, Indiana*

Purdue University **August 2023 – May 2027**
B.S. in Computer Science & Data Science, Minor in Math (3.99 GPA) *West Lafayette, Indiana*

Experience

Amazon Web Services (AWS) **May 2026 – August 2026**
Cloud Consultant Intern *Arlington, Virginia*

- Designed and developed a cloud-native geospatial tile delivery API leveraging Amazon ECS Fargate, Amazon S3, and Amazon Aurora to serve over **1 billion** detections with sub-second response times for **100+** concurrent users
- Engineered an AI-powered analytics interface to monitor API usage and analyze overall system performance
- Developed a pipeline that regenerates map tiles as detections enter Amazon Aurora, enabling real-time API data

Cummins Inc. **July 2025 – December 2025**
Software Engineering Intern *West Lafayette, Indiana*

- Developed a company-specific transformer in Python to interpret customer messages and retrieve relevant products
- Created auditing dashboards and automations in Jira to streamline intake workflows and improve efficiency by **40%**
- Conducted audits and developed internal company pages to enhance usability and improve accessibility of resources

Indiana State Department of Health **May 2025 – July 2025**
Data Science Intern *Indianapolis, Indiana*

- Created address lookup system to automate **150,000+** data entries with an Apache server containerized using Docker
- Used Selenium to enter thousands of lab details into the National Electronic Disease Surveillance System (NEDSS)

Texavi **June 2024 – August 2024**
Software Development Intern *Remote*

- Built a dynamic WordPress CMS for a UK archery organization to support easy updates and user engagement
- Improved website structure, layout, and navigation to enhance overall usability, accessibility, and user experience
- Implemented email management strategies to streamline communication with customers and improve outreach efficiency

Projects

AI-Driven Nurse Assistant | *React Native, FastAPI, LangChain, PostgreSQL* **August 2025 – December 2026**

- Developed a dynamic NLP pipeline using LangChain and Ollama with Pydantic-based parsing and GPU acceleration to efficiently extract structured sentiment data from nurse feedback and generate actionable managerial recommendations
- Designed analytics and prediction logic to model nurse supply and demand, converting unstructured Teams text data under varying levels of model context into real-time availability indicators to support informed staffing decisions
- Developed a TypeScript frontend with Microsoft Teams API to analyze nurse feedback and workforce patterns

Tradify | *Alpaca, PostgreSQL, WebSockets, Matplotlib* **August 2025 – December 2025**

- Implementing a multi-component trading pipeline, including a continuously running data collector to fetch live price, news, and sentiment data from APIs, store it in a SQL database, and update an in-memory cache for fast access
- Developing a signal generator and trading engine to generate buy/sell/hold signals based on a custom-trained ML model, apply risk management rules, and execute trades via Alpaca, while logging trade results and updating portfolio status

ReciPic: AI-Powered Recipe Finder | *React, Pytorch, Flask, Firebase, Gemini* **April 2025**

- Built an ingredient-recognition pipeline using computer vision to identify food items from user-uploaded images using a custom-trained YOLOv8 model, storing the outputs to Firebase Storage
- Integrated Gemini to produce recipes and infer additional needed ingredients from partial fridge inventories
- Combined ML outputs with Walmart's platform to auto-purchase missing ingredients through a Chrome extension

Ports | *Python, Java, Maven* **Februrary 2025 – May 2025**

- Researched portfolio risk calculation methods for Ports, a fintech startup for investing in user-generated portfolios
- Developed a risk quantifier and implemented tests to optimize portfolio performance using the Central Market Line
- Built a model incorporating gradient-descent logic to maximize the Sharpe ratio, integrating additional risk measures

Technical Skills

Languages: Python, Java, C++, C, HTML/CSS, Javascript, GDScript, R, Typescript
Technologies/Frameworks: ReactJS, NextJS, React Native, Tailwind, Pytorch, Bootstrap, GitHub, Linux, Maven, AWS
Certifications: GIAC Foundational Cybersecurity Technologies, Akuna Options 101, AWS Certified Cloud Practitioner, AWS Certified AI Practitioner, AWS Certified Machine Learning Engineer, AWS Certified Solutions Architect Associate